On Existence of Digital Measure Constant

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***Abstract*—Unidentified continuous functions are a generalized classification for all signals. The author assumes that any unidentified continuous function (any signal) is either digital (non-Archimedean continuum) or analog (Archimedean continuum), and gives a method to indicate a distinction between both continuum models. Following Luzin's saw, the digital model is represented by ceil/floor series, whose inclination is horizontal on every step and vertical between steps, at every scale. The distinction is measured by the digital measure constant Hyp_1, the difference in length between the two models that this difference in inclination causes. We show that $Hyp_1 = \int(1+|f'|-\sqrt{1+f'^2})\,dx$, that it is zero only for constant functions, and that it does not depend on the step, so it does not vanish at large scale, unlike the area difference. The length ratio of the two models always lies between 1 and the universal constant $\sqrt{2}$, which is reached exactly at inclination ± 1. The method can therefore be applied to all unidentified continuous functions in order to identify their continuum models.**

***Index Terms*—Time-frequency analysis, non-standard analysis, digital topology**

I. Introduction

THE article "Approaches to analysis with infinitesimals following Robinson, Nelson and others" [18] has the section "The CD+II mindset". The terms of that section will be regarded, since they are foundations of set-theoretical identifications that are based on an "intuitively equal" algebraic partition (arithmetization) of the real number line (all real numbers) into a countably infinite amount of interval groups. (The work of Sophus Lie is a general reference for the historical viewpoint of continuous symmetric groups [2]–[4].)

"The Cantor–Dedekind postulate (CD) identifying the line in physical space with the real number line" (Cantor's main work is [5]; his arguments on reals and infinitesimals are on pages 26–55 of [17]).

"The Intended Interpretation hypothesis (II), entailing an identification of a standard $\mathbb{N}$ in its set-theoretic context, on the one hand, with ordinary intuitive counting numbers, on the other."

These main arguments will be the building blocks for the separation of unidentified continuous functions (generalized signals). The mathematical foundations are generally regarded whenever there is a need for deductive (Fourier or harmonic) functional analysis (see [20]). The author will regard non-standard analysis, since the main arguments for a digital continuum can be founded via inductive (instantaneous) non-standard notions of analysis. Rather than concerning applications about the statistical large-scale construction of signals on the real numbers to generate a generalized theory, the author establishes a deterministic model of small-scale construction for signals on $\mathbb{R}^*$ as a generalized theory and applies it to larger scales.

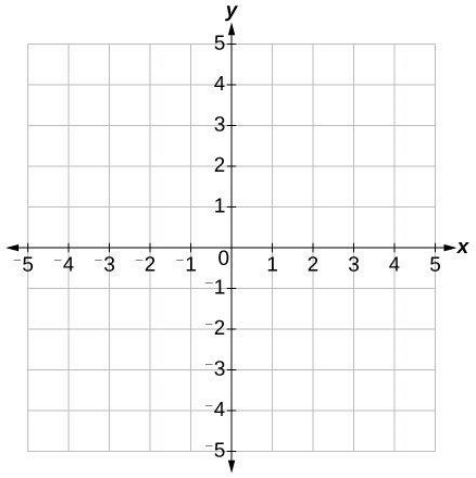


Fig. 1. Cartesian Coordinate System

The Cartesian Coordinate System is mainly a product of the algebraic partition of the real number line into "intuitively equal", countable and infinitely many interval parts. The branches of mathematics, mainly set theory, measure theory and analysis, define different terms regarding this application of partition. The author adds another term, the "Coarsity of Compression Model Hypothesis" (CCMH). This hypothesis is the assumption that all (a priori) proportionately partitioned parts can always be partitioned proportionately again, as a generalization. In order to explain the CCMH, the author will regard types of continuum and the general view of scales as perspectives. Finally, the author will define the digital measure constant as a continuum relation.

While the Cartesian Coordinate System is the human mind's intuition to partition physical space into patterns in order to recognize physical space itself, the term "intuitively equal" is used rather than "equal" because the real number set (the real number line) is uncountable in terms of counting numbers. Via non-standard analysis one can construct the same a priori idea of partition on the unit interval, and then apply this idea of partition to unidentified continuous functions. Work of this type has been done before in Physics Letters A by R. F. Werner and M. P. H. Wolff [21].

Revised version, 31 October 2026. This version keeps the original argument and states its results in precise form: Hyp_1 as the inclination (length) difference between the two continuum models, the universal bound $\sqrt{2}$, the unit relation behind Hyp_0, and the logical energy unit u.

A. Types of Continuum for Unidentified Continuity

We begin with the idea of certain graphs (images) which look identical when looked at from far away. Since a "function" is the representation of an image on a graph, the distinctions between small-scale models of *"functions that look identical from far away"* will be separated into two kinds of continuum. This analysis was made in the article *"Tension between Intuitive Infinitesimals and Formal Mathematical Analysis"* [19] by Mikhail G. Katz and David Tall. The author mainly changed Luzin's saw (sawtooth) into ceil/floor series as a generalization.

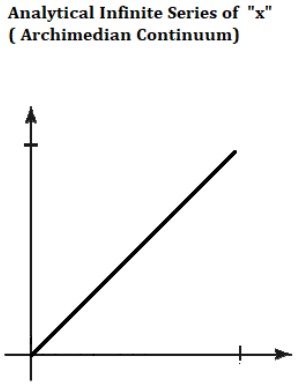


Fig. 2. Analog Continuum Model as x

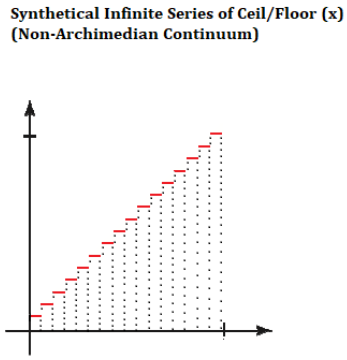


Fig. 3. Digital Continuum Model as $\lceil x \rceil$

In Figures 2–3, the two graphs (images) of functions look identical when looked at from a far distance. For the generalization, the term "unidentified" will be used for functions f (see below) that look like generalizations of an infinitely differentiable function (n indicates infinite) when looked at from far away (large scale), but are "unknown" in terms of continuum (small scale):

$$\frac{\partial^n f}{\partial x^n} \cong 0\,; \qquad f(x) \cong \sum_{i\in\mathbb{N}} b_i \cdot x^i\,; \quad b_i \in \mathbb{R}$$

Here and below, $A \cong B$ means that $A - B$ is infinitesimal, that is, A and B are equal when looked at from large scale.

Any area below the curve of the two possible continuum models of an "unidentified" function will also look like the same constant number when looked at from far away. But in reality there will be a difference between the defined continuum models: their inclination. However small the steps become, the digital model keeps a horizontal inclination on each step and a vertical one between steps, just as the teeth of Luzin's saw keep their inclination while the saw itself flattens.

The reason why ceil/floor series are used for the digital (non-Archimedean) continuum model, rather than the functional representations described in *"Tension between Intuitive Infinitesimals and Formal Mathematical Analysis"* at pages 14–15, is that the *"Natural Number Set"* representation as *"intuitively equal"* interval partitions can be shown with ceil/floor series.

II. Generalized Functional Model Interpretations

The abstraction is for the functional integration of natural numbers (as groups) with real numbers. Any continuum model of an "unidentified" function f at small scale is therefore shown with these functional representations. For a step $dx > 0$, write the ceil and floor at that step as

$$\lceil x \rceil_{dx} = dx \cdot \left\lceil \frac{x}{dx} \right\rceil, \qquad \lfloor x \rfloor_{dx} = dx \cdot \left\lfloor \frac{x}{dx} \right\rfloor$$

The author will regard the ceil and floor series as the digital models of f at step dx. The coefficients a_i are natural numbers here, so f is increasing for $x \geq 0$. The ceil and floor series do not represent f through area but through inclination: on each step the digital model is horizontal (inclination 0), and between steps it rises vertically. The length element of the digital model is therefore the sum of its horizontal and vertical parts, while the length element of the analytical model follows the inclination f' of the curve:

$$ds_D = dx + |df|\,, \qquad ds_A = \sqrt{dx^2 + df^2} = \sqrt{1 + f'(x)^2}\,dx$$

The constant Hyp_1 is the representation of the inclination difference between the models, measured as the difference of their lengths. Writing $L(g)$ for the length of the graph of g on $[0, H]$, with the vertical risers of a step function included,

$$L\Big(\sum_{i\in\mathbb{N}} a_i \lceil x \rceil_{dx}^{i}\Big) - L\Big(\sum_{i\in\mathbb{N}} a_i x^i\Big) = Hyp_1$$

$$L\Big(\sum_{i\in\mathbb{N}} a_i \lfloor x \rfloor_{dx}^{i}\Big) - L\Big(\sum_{i\in\mathbb{N}} a_i x^i\Big) = Hyp_1$$

Both equations hold exactly and at every step dx: the ceil and floor staircases consist of the same horizontal steps and the same vertical risers (Proposition 1).

Note: The integration is taken on the domain $[0, H]$ because the generalized functional model is applied from a small-scale perspective. Any definite integration of length, area or volume that

results in a constant number from the large-scale perspective is assumed to be "potentially infinite" in small-scale models. Here H is such a potential infinity: an infinite hyperreal bound, so that the integral is a well-defined element of $\mathbb{R}^*$. Hence the representation of the digital continuum is possible regardless of the basis domain. The term "actual infinite" (∞) will be used as the "singular infinite", and "potential infinite" (H) as a perspective.

III. Model Interpretations on Small Scale

The abstraction is to complement the smaller-scale perspective. The symbol $(dx)_{\|\mu\|}$ will indicate the digital (non-Archimedean) continuum at small-scale perspective. The digital continuum $(dx)_{\|\mu\|}$ has an interrelation with ceil/floor series as a functional representation. This is mainly indicated via the hyperreal set, shown with the symbol $\mathbb{R}^*$. (Here the author must cite Hilbert [8]–[10], Hausdorff [6], [7], Lebesgue [11], Borel [12], [13] and Robinson [15], because of various foundations and relations in analysis, set theory and measures.)

$$0 < a < +\infty\,; \qquad a = n \cdot (dx)_{\|\mu\|}$$

$$a \in \mathbb{R}^*\,; \quad n \in \mathbb{N}^+\,; \qquad \frac{\pm 1}{(dx)_{\|\mu\|}} \triangleq \pm\infty$$

Here the non-integer coefficient variants of $(dx)_{\|\mu\|}$ unit/group lengths are non-definitive for the $(dx)_{\|\mu\|}$ units/groups. So the CCMH is basically an "Intended Interpretation" model on the unit interval.

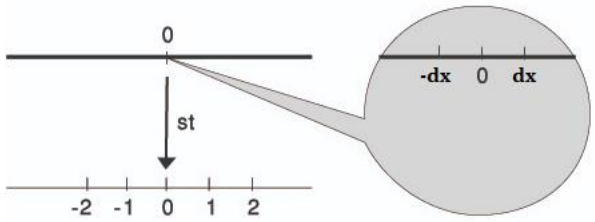


Fig. 4. A proportionate partitioning

IV. Digital Measure Constant

On earlier pages the statement *"Any area below the curve of the two possible continuum models of an unidentified function will also look like the same constant number when looked at from far away. But in reality there will be a difference between the defined continuum models"* was given. By area, this difference vanishes at large scale; by inclination, it does not. The author will regard the type of "unidentified" function f with the term

$$\Delta x \cdot \int_{\Omega_1} f$$

for functions that are variations of analytical models, which have an Archimedean continuum at small scale. The author will also regard the type of "unidentified" function f from a closer view with the term

$$\int_{\Omega_1} f\,dx$$

for functions that are variations of synthetical models, which have a digital (non-Archimedean) continuum at small scale. Precisely, $\Delta x \cdot \int_{\Omega_1} f$ is the integral of the analytical model, and $\int_{\Omega_1} f\,dx$ is the step sum of the synthetical model, $\sum_k f(\xi_k)\,dx$, taken over cells of length dx with $f(\xi_k)$ the upper (or lower) value of f on each cell.

Since any finite area below the curve of an "unidentified" function in the large-scale view looks non-finite when looked at from the small-scale view, the area difference between the models can be shown as nearly equal to zero in the large-scale view.

If on $\Omega_1 := [a,b]$ the large-scale integration of f is known to be a finite constant S,

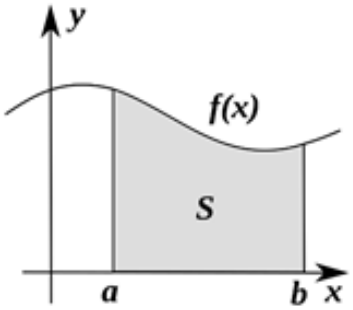


Fig. 5. Area under the curve of unidentified function

then the equations below must hold for the large-scale view:

$$\Delta x \cdot \int_{\Omega_1} f \cong S\,, \qquad \int_{\Omega_1} f\,dx \cong S\,, \qquad S \in \mathbb{R}$$

$$\left|\Delta x \cdot \int_{\Omega_1} f - \int_{\Omega_1} f\,dx\right| \cong 0$$

Note: If the domain or range of the function is extended to infinity anyhow, it must be extended to infinity regardless of all perspectives. Here the functional representation of "proportion" is entirely due to the perspective; therefore a potential point resides *at some infinity, not "the infinity" itself* [14]. This is due to Carathéodory's extension theorem (lemma):

$$\mu^*(S) = \mu^*(S \cap M) + \mu^*(S \cap M^c)$$

However, the two continuum models are not distinguished by area. They are distinguished by inclination: the digital model is horizontal on every step and vertical on every riser, at every scale, while the analytical model follows the inclination f'. This difference does not shrink with the step and does not vanish at large scale, also when the integration of f is non-finite. Hence

the difference between the models results in the digital measure constant, measured by length:

$$Hyp_1 = L_D - L_A$$

where L_D and L_A are the lengths of the digital and analytical models on Ω_1. Hyp_1 is a constant because it determines the measurability of the two models. It cannot be equal to zero, since that would mean the two continuum models are equal. And it must be a constant, since if the difference between the two possible continuum models did not hold as a constant, definite (finite) measurements in the large-scale view would give different results for the two models because of the expansion of the compared maps (graphs, images), which would contradict large-scale definite integration (Gromov [16]). The value of this constant is given precisely as follows.

Proposition 1: Let f be continuously differentiable on $\Omega_1 = [a,b]$, and let the digital model be the ceil (or floor) staircase of f at step dx, on a grid containing a and b. Then

$$L_D = (b-a) + V_{dx}(f), \qquad L_A = \int_a^b \sqrt{1+f'(x)^2}\,dx$$

where $V_{dx}(f)$ is the sum of the riser heights. For monotone f, $V_{dx}(f) = |f(b) - f(a)|$ for every step dx; in general $V_{dx}(f) \to V_a^b(f) = \int_a^b |f'(x)|\,dx$. Hence

$$Hyp_1 = \int_a^b \left(1 + |f'(x)| - \sqrt{1+f'(x)^2}\right) dx$$

exactly for monotone f at every step, and in the limit $dx \to 0$ otherwise.

Proof: The horizontal steps have total length $b-a$. The risers have heights $|f(x_{k+1}) - f(x_k)|$, whose sum telescopes to $|f(b) - f(a)|$ for monotone f and converges to the total variation for continuously differentiable f. The length of the analytical model is the arc-length integral. Subtracting gives Hyp_1. ■

Proposition 1 gives the claimed properties of Hyp_1 exactly. Write $g(s) = 1 + |s| - \sqrt{1+s^2}$ for the integrand at inclination s; then $0 \le g(s) < 1$, with $g(s) = 0$ only at $s = 0$.

1) *Non-zero:* $Hyp_1 = 0$ only when $f' \equiv 0$, that is, when f is constant and the two continuum models coincide. For every other function, $Hyp_1 > 0$.
2) *Constant at every scale:* Hyp_1 does not depend on the step dx. In $\mathbb{R}^*$, with infinitesimal dx, the digital model is infinitely close to f in value, yet its length is not infinitely close to the length of f. So $Hyp_1 \ncong 0$: the distinction survives at large scale.
3) *Ceil and floor agree:* both staircases have the same steps and risers, so both equations of Section II hold exactly.
4) *Universal constant:* at inclination s the ratio of the two length elements is $(1+|s|)/\sqrt{1+s^2}$, which is at most $\sqrt{2}$, with equality exactly at $|s| = 1$. Hence, for every function, every step and every scale,

$$1 \le \frac{L_D}{L_A} \le \sqrt{2}$$

The bound $\sqrt{2}$ is reached exactly at inclination ± 1, the inclination of the teeth of Luzin's saw. The ratio is unchanged when both axes are rescaled together, so $\sqrt{2}$ is a dimensionless universal constant.

For example, for $f(x) = x$ on $[0,1]$, $L_D = 2$ and $L_A = \sqrt{2}$, so $Hyp_1 = 2 - \sqrt{2} \approx 0.586$ and the ratio is exactly $\sqrt{2}$. For $f(x) = x^2$ on $[0,1]$, $L_D = 2$, $L_A \approx 1.479$ and $Hyp_1 \approx 0.521$. Since $g(s) < 1$, $Hyp_1 < b - a$ for every function.

Remark (area): By area, the two models do become indistinguishable. The area difference between the ceil (or floor) model and f is $\frac{1}{2}\,dx \cdot V_a^b(f) + o(dx)$, which is $\cong 0$ at large scale. Area and value cannot separate the two continuum models from far away; inclination always can. This is the method to indicate the distinction.

The two perspectives can also be compared through the integral itself. There is a special case when f is dx. This is because dx acts as a *constant* on $\mathbb{R}$, because $\mathbb{R}$ is *Archimedean*, and acts as an *infinitesimal* on $\mathbb{R}^*$, because it is *non-Archimedean*:

$$f(x) \cong dx$$

The integral therefore has to abide by two situations. The first is when f acts as the *constant number* $dx > 0$, with the actual infinity as upper bound:

$$\int_0^\infty dx \cdot dx = dx \cdot \int_0^\infty dx = \infty$$

The equations above hold since the large-scale integration of this integral is infinite. The second is when f acts as the *infinitesimal number* dx. The upper bound is then the potential infinity of the small-scale perspective, $H = 1/dx$:

$$\frac{\pm 1}{(dx)_{\|\mu\|}} \triangleq \pm\infty$$

$$\int_0^H dx = \int_0^{1/dx} dx = \frac{1}{dx} = 1 + 1 + 1 + \ldots$$

where the sum has $1/dx$ terms. Therefore

$$\int_0^H dx \cdot dx = dx \cdot \int_0^{1/dx} dx = 1$$

The two situations use different upper bounds, the actual infinity ∞ and the potential infinity $H = 1/dx$, which is why they give different values. This special case does not measure the inclination difference of Proposition 1: the constant function dx has inclination 0 in both models, so $Hyp_1 = 0$ for it. Instead it compares the two perspectives. In the analytical perspective one unit of length is Δx; in the small-scale perspective the same unit is $dx \cdot (1/dx)$ steps. Their ratio defines the second constant:

$$\frac{\Delta x}{dx} = Hyp_0$$

that is,

$$\Delta x = Hyp_0 \cdot dx$$

Meaning: any unit distance Δx on $\mathbb{R}$ contains Hyp_0 steps of the digital continuum $\mathbb{R}^*$. Hyp_0 is the conversion factor between the two units, like the number of centimetres in a metre. With $\Delta x = 1$, $Hyp_0 = 1/dx$ is an infinite hypernatural number. In order to obtain the values of the digital measure constants Hyp_0 and Hyp_1, the author indicates that the generalized functional model interpretations of Section II lead to the result: the inclination (length) difference between the ceil/floor and analytic models with natural coefficients is the Hyp_1 constant, whose value is given by Proposition 1.

V. Digital Measure

Any function that represents a generalized signal and has unidentified continuity shown on the real numbers $\mathbb{R}$ (far-away view) can also be shown on the hyperreal number set $\mathbb{R}^*$. The author indicates the existence of another digital measure constant, Hyp_0, via generalization, in order to show that a point at real coordinate x lies at digital coordinate (step count) x^* with

$$x^* = Hyp_0 \cdot x \; ; \qquad x \in \mathbb{R} \; ; \quad x^* \in \mathbb{R}^*$$

The digital measure of an interval $[a,b]$ on $\mathbb{R}$ is its number of steps:

$$\mu_D[a,b] = Hyp_0 \cdot (b-a) \; ; \qquad b > a$$

Equivalently, an interval $[a^*,b^*]$ given in digital coordinates has real length

$$\frac{1}{Hyp_0} \cdot (b^* - a^*)$$

VI. Applications

If there is a signal f (shown via a function) that is known to have unidentified continuity,

$$\frac{\partial^n f}{\partial x^n} \cong 0 \; ; \qquad f(x) \cong \sum_{i \in \mathbb{N}} b_i \cdot x^i \; ; \quad b_i \in \mathbb{R}$$

then via this signal one can obtain the digital continuum model or the analog continuum model of this signal. So any unidentified continuous function f can be transferred from one continuum model to the other by

$$f(x) = f\left(\frac{x^*}{Hyp_0}\right) \; ; \qquad x \in \mathbb{R} \; ; \quad x^* \in \mathbb{R}^*$$

An example is the energy of the photon (wave–particle), shown via functions ($E(x)$, $f(x)$ and $\lambda(x)$ for the functional representations of energy, frequency and wavelength) that have unidentified continuity:

$$E(x) = h \cdot f(x) \; ; \qquad f(x) = \frac{c}{\lambda(x)} \; ; \qquad \lambda(x) = x$$

Then the digital continuum model of wavelength (as an interval) is

$$\lambda = \frac{\lambda^*}{Hyp_0} \; ; \qquad \lambda^* \in \mathbb{R}^* \; ; \quad \lambda \in \mathbb{R}$$

with the bounded range

$$\lambda^* \in \mathbb{R}^* \; ; \qquad 0 \le \lambda^* \le \frac{1}{dx}$$

If every wavelength is an integer multiple of the interval $[0,dx]$, then $\lambda^* = n$ with $n \in \mathbb{N}^+$ and $n \le N$, where $N = 1/dx$. The energies are then

$$E_n = h \cdot \frac{c}{\lambda} = \frac{h\,c\,Hyp_0}{n} \; ; \qquad n = 1,2,\ldots,N$$

These energy levels are not integer multiples of each other; they form the sequence $E_1, E_1/2, E_1/3, \ldots$ Nevertheless, a single *logical* unit u exists such that every level is an integer multiple of it. Let $L = \mathrm{lcm}(1,2,\ldots,N)$, which exists as a hypernatural number by transfer. Then

$$u = \frac{h\,c\,Hyp_0}{L} \; ; \qquad E_n = \frac{L}{n} \cdot u \; , \quad \frac{L}{n} \in \mathbb{N}^+$$

For example, for a finite range with $N = 10$, $L = 2520$ and the levels are $2520u, 1260u, 840u, \ldots$ The bounded range of λ^* is essential: without an upper bound on n, no single unit divides every level. The author therefore indicates that if every wavelength in a bounded range is an integer multiple of the interval $[0,dx]$, then every energy level is an integer multiple of the logical constant u. This holds within the digital continuum model and is not a statement about measured photon spectra.

VII. Conclusion

Within the non-Archimedean continuum model, the digital measure constant exists. The ceil/floor (digital) model and the analytical model look identical from far away in value and in area: their area difference is $\frac{1}{2}\,dx \cdot V_a^b(f)$, which vanishes at large scale. They never look identical in inclination: $Hyp_1 = \int (1 + |f'| - \sqrt{1+f'^2})\,dx$ is nonzero for every non-constant function and the same at every scale, and the length ratio of the two models is bounded by the universal constant $\sqrt{2}$, reached at inclination ± 1 as in Luzin's saw. The constant $Hyp_0 = \Delta x/dx$ converts between the real and the digital units of length.

Data Availability Statement

Some of the data is available to the public, while some of the data has prerequisites, and some of the data is located at an outdated website. The author has given information about the contents he has read below. All referenced pages are at the equivalent of the internet domain's reference pages (logical), not the original book's or reference's pages (physical).

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